Experiment 15: Probabilistic Context-Free Grammar Parsing in Python

1. Aim

To write a Python program to implement probabilistic context-free grammar (PCFG) parsing for a sentence.

2. Procedure

1. Import `PCFG` and `ViterbiParser` from the `nltk` module.
2. Define a probabilistic grammar with rule probabilities using `PCFG.fromstring()`.
3. Create a parser object using `nltk.ViterbiParser()`.
4. Parse the tokenized sentence to compute the most probable parse tree.
5. Display the resulting probabilistic parse tree and its probability score.

3. Program

```
import nltk
from nltk import PCFG

pcfg = PCFG.fromstring("S -> NP VP [1.0]\nNP -> Det N [0.6] | 'John' [0.4]\nVP -> V NP [1.0]\nDet -> 'the' [1.0]\nN -> 'dog' [1.0]\nV -> 'saw' [1.0]")

parser = nltk.ViterbiParser(pcfg)
sentence = "John saw the dog".split()

for tree in parser.parse(sentence):
    print("PCFG Parse:", tree)
```

4. Output

```
PCFG Parse: (S (NP John) (VP (V saw) (NP (Det the) (N dog)))) (p=0.24)
```

5. Result

Thus, the Python program was successfully executed to implement probabilistic context-free grammar parsing using `ViterbiParser`.